Assignment - 4

~~(a) (1) $\sigma_+ = \frac{\sigma_x + \sigma_y}{2}$; $\sigma_- = i(\sigma_x - \sigma_y)$~~

(a) $\sigma_x = \sigma_+ + \sigma_-$; $\sigma_y = i(\sigma_- - \sigma_+)$

$\therefore \sigma_x^1 \sigma_x^2 + \sigma_y^1 \sigma_y^2$

$= (\sigma_+^1 + \sigma_-^1)(\sigma_+^2 + \sigma_-^2) - (\sigma_-^1 - \sigma_+^1)(\sigma_-^2 - \sigma_+^2)$

$= \cancel{\sigma_+^1 \sigma_+^2} + \sigma_+^1 \sigma_-^2 + \sigma_-^1 \sigma_+^2 + \cancel{\sigma_-^1 \sigma_-^2}$
$- (\cancel{\sigma_-^1 \sigma_-^2} - \sigma_-^1 \sigma_+^2 - \sigma_+^1 \sigma_-^2 + \cancel{\sigma_+^1 \sigma_+^2})$

$= 2(\sigma_+^1 \sigma_-^2 + \sigma_-^1 \sigma_+^2)$

$\therefore H = -2J(\sigma_+^1 \sigma_-^2 + \sigma_-^1 \sigma_+^2)$

The ~~eigenbasis~~ states for $2-e^-$ system:

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$$H|\uparrow\uparrow\rangle = -2J(\sigma_+^1\sigma_-^2 + \sigma_-^1\sigma_+^2)|\uparrow\uparrow\rangle = 0$$

$$H|\downarrow\downarrow\rangle = 0.$$

$$H|\uparrow\downarrow\rangle = -2J(\sigma_+^1\sigma_-^2 + \sigma_-^1\sigma_+^2)|\uparrow\downarrow\rangle = -2J.|\downarrow\uparrow\rangle$$

$$H|\downarrow\uparrow\rangle = -2J(\sigma_+^1\sigma_-^2 + \sigma_-^1\sigma_+^2)|\downarrow\uparrow\rangle = -2J|\uparrow\downarrow\rangle$$

$$\therefore \quad H = \begin{pmatrix} 0 & -2J \\ -2J & 0 \end{pmatrix} \quad [\text{ in } |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle \text{ basis}]$$

Let c be the eigenvalue of H.

$$|H - c\mathbb{I}| = 0$$

$$\Rightarrow \begin{vmatrix} -c & -2J \\ -2J & -c \end{vmatrix} = 0 \quad \Rightarrow \quad c^2 - (2J)^2 = 0$$

$$\Rightarrow \quad \boxed{c = \pm 2J}$$

$\underline{c = 2J}$:

Let the ~~wave~~ eigenstate be $|\varphi_+\rangle = A_+\begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix}$ $\left[\cancel{\varphi_1 = |\uparrow\downarrow\rangle, \varphi_2 = |\downarrow\uparrow\rangle}\right]$

$$\therefore \quad H|\varphi_+\rangle = 2J|\varphi_+\rangle \qquad \left[\begin{array}{l} |\uparrow\downarrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |\downarrow\uparrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{array}\right]$$

$$\Rightarrow \quad (H - 2J)|\varphi_+\rangle = 0$$

$$\Rightarrow \quad -2JA_+\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \varphi_1 \\ \varphi_2 \end{pmatrix} = 0 \quad \Rightarrow \quad \varphi_1 + \varphi_2 = 0$$

$$\Rightarrow \quad \varphi_1 = -\varphi_2$$

$$\therefore \quad |\varphi_+\rangle = A_+\begin{pmatrix} 1 \\ -1 \end{pmatrix} = A_+[|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle]$$

$$\therefore \quad \langle\varphi_+|\varphi_+\rangle = 1 \quad \Rightarrow \quad 2|A_+|^2 = 1 \quad \Rightarrow \quad \boxed{|A_+| = \frac{1}{\sqrt{2}}}$$

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$\therefore \quad |\psi_+\rangle = \frac{1}{\sqrt{2}} [\, |\uparrow\downarrow\rangle \mp |\downarrow\uparrow\rangle \,]$.

$\underline{c = -2J.}$ 1

$$|\psi_-\rangle = A_- \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}.$$

$$\therefore \quad A_- \begin{pmatrix} 2J & -2J \\ -2J & 2J \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = 0.$$

$$\Rightarrow \quad \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = 0 \quad \Rightarrow \quad \psi_1 = \psi_2$$

$$\therefore \quad |\psi_-\rangle = A_- \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$ (from normalisat^n)

$$|\psi_-\rangle = \frac{1}{\sqrt{2}} [\, |\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle \,]$$

State	Energy	Degeneracy
$\|\uparrow\uparrow\rangle, \|\downarrow\downarrow\rangle$	0	2
$\frac{1}{\sqrt{2}}(\|\uparrow\downarrow\rangle + \|\downarrow\uparrow\rangle)$	$-2J$	1
$\frac{1}{\sqrt{2}}(\|\uparrow\downarrow\rangle - \|\downarrow\uparrow\rangle)$	$2J$	1

(b) $\hat{H}_z = -\frac{g\mu_B B}{2} (\sigma_z^1 + \sigma_z^2) = -h(\sigma_z^1 + \sigma_z^2)$

$\therefore \; H_z |\uparrow\uparrow\rangle = -2h|\uparrow\uparrow\rangle; \quad H_z |\downarrow\downarrow\rangle = 2h\,|\downarrow\downarrow\rangle$

$H_z |\uparrow\downarrow\rangle = 0 = H_z |\downarrow\uparrow\rangle$

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$$\therefore \hat{H}_z = \begin{pmatrix} -2h & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2h \end{pmatrix}$$ [in $|\uparrow\uparrow\rangle$, $|\downarrow\downarrow\rangle$, $|\uparrow\downarrow\rangle$, $|\downarrow\uparrow\rangle$ basis]

$$H_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -2J & 0 \\ 0 & -2J & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$ ["]

$$\therefore \hat{H}_{total} = \begin{pmatrix} -2h & 0 & 0 & 0 \\ 0 & 0 & -2J & 0 \\ 0 & -2J & 0 & 0 \\ 0 & 0 & 0 & 2h \end{pmatrix}$$

$$H_{total} = H_0 + H_z$$

$$\therefore H_{total}|\uparrow\uparrow\rangle = H_0|\uparrow\uparrow\rangle + H_z|\uparrow\uparrow\rangle = -2h|\uparrow\uparrow\rangle$$

$$H_{total}|\downarrow\downarrow\rangle = H_0|\downarrow\downarrow\rangle + H_z|\downarrow\downarrow\rangle = 2h|\downarrow\downarrow\rangle$$

$$H_t|\uparrow\downarrow\rangle = -2J|\uparrow\downarrow\rangle \; ; \; H_t|\downarrow\uparrow\rangle = -2J|\downarrow\uparrow\rangle$$

$|\uparrow\downarrow\rangle$ & $|\downarrow\uparrow\rangle$ are not the eigenstates of H_t

$$H_z|\psi_+\rangle = 0 = H_z|\psi_-\rangle$$

$\therefore$ $|\psi_+\rangle$ & $|\psi_-\rangle$ are eigenstates of H_t.

$$H_t|\psi_+\rangle = 2J|\psi_+\rangle \; ; \; H_t|\psi_-\rangle = -2J|\psi_-\rangle$$

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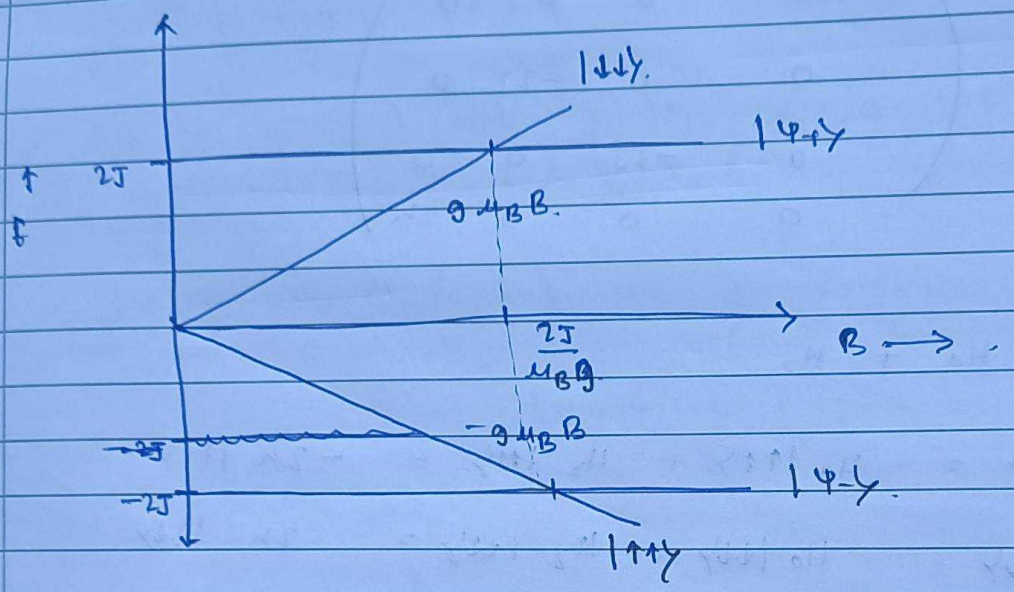
State	Energy
$\lvert\uparrow\uparrow\rangle$	$-2h = -g\mu_B B$
$\lvert\downarrow\downarrow\rangle$	$2h = g\mu_B B$
$\lvert\psi_+\rangle$	$2J$
$\lvert\psi_-\rangle$	$-2J$



② (a) $H = \frac{L^2}{2I} = -\frac{\hbar^2}{2I}\frac{d^2}{d\phi^2}$

$\therefore$ ~~$H\phi = \lambda\phi$~~ $\quad H\psi = \lambda\psi$.

~~$\Rightarrow -\frac{\hbar^2}{2I}\phi'' = \lambda\phi$~~ $\Rightarrow -\frac{\hbar^2}{2I}\psi'' = \lambda\psi$.

~~$\Rightarrow \phi''$~~ $\Rightarrow \psi'' + \frac{2I\lambda}{\hbar^2}\psi = 0$

$\boxed{k^2 = \frac{2I\lambda}{\hbar^2}}$ $\Rightarrow \boxed{\psi'' + k^2\psi = 0}$

General Sol^n: $\cancel{Ae}\ \psi(\phi) = Ae^{ik\phi} + Be^{-ik\phi}$.

B.C.s: $\psi(\phi) = \psi(\phi+2\pi)$.

$\Rightarrow Ae^{ik\phi} + Be^{-ik\phi} = Ae^{ik\phi} \cdot e^{ik2\pi} + Be^{-ik\phi} \cdot e^{-ik2\pi}$

$\therefore e^{ik2\pi} = 1 \Rightarrow k \in \mathbb{Z} = m$.

$e^{-ik2\pi} = 1 \Rightarrow e^{ik2\pi} = 1 \Rightarrow k \in \mathbb{Z} = m$.

$\therefore \psi(\phi) = Ae^{ik\phi} = Ae^{im\phi}$ [we choose only $e^{im\phi}$ term as the other term appears automatically for $m<0$]

Normalisat^n:

$$\int_0^{2\pi} |\psi(\phi)|^2 d\phi = 1 \Rightarrow |A|^2 \int_0^{2\pi} d\phi = 1$$

$$\Rightarrow \boxed{|A| = \frac{1}{\sqrt{2\pi}}}$$

$$\therefore \boxed{\psi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}} \quad m \in \mathbb{Z}$$

The quantum no. m characterises $\psi(\phi)$.

(b) Energy eigenvalues: λ =

$$\cancel{\frac{L^2\psi(\phi)}{2I}}\ H\psi(\phi) = \frac{L^2}{2I}\psi(\phi) = -\frac{\hbar^2}{2I}\frac{d^2}{d\phi^2} \cdot \frac{1}{\sqrt{2\pi}} e^{im\phi}$$

$$= -\frac{\hbar^2}{2I} \cdot -m^2 \psi(\phi)$$

$$\therefore \boxed{\lambda = \frac{m^2\hbar^2}{2I}} \text{ (Ans)}$$

$$\boxed{H\psi(\phi) = \frac{m^2\hbar^2}{2I}\psi(\phi)}$$

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(c) $\mathrm{Re}\,[\psi(\phi)] = \frac{1}{\sqrt{2\pi}} \cos m\phi$.

Ground state : $m=0 \longrightarrow \psi_0 = \frac{1}{\sqrt{2\pi}}$

1st excited state : $m=1 \rightarrow \psi_1 = \frac{\cos\phi}{\sqrt{2\pi}}$

2nd excited state : $m=2 \rightarrow \psi_2 = \frac{\cos 2\phi}{\sqrt{2\pi}}$

(d) $H = \frac{\hbar^2}{2I}\left(-i\frac{d}{d\phi} - \alpha\frac{\phi}{\phi_0}\right)^2 \cancel{= \frac{\hbar^2}{2I}\left(i\frac{d}{d\phi} + \frac{\alpha\phi}{\phi_0}\right)^2}$

$\cancel{= \frac{\hbar^2}{2I}\left(-\frac{d^2}{d\phi^2} + 2\alpha i\frac{d}{d\phi} + \alpha^2\right)}$ $\left[\alpha = \frac{\phi}{\phi_0}\right]$

$\cancel{= H_0}$

Let us try the same $\psi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{ik\phi}$

$\therefore \quad -i\frac{d}{d\phi}\psi = k\psi.$

$\therefore \quad \left(-i\frac{d}{d\phi} - \alpha\right)\psi = (k-\alpha)\psi.$

$\therefore \quad \left(-i\frac{d}{d\phi} - \alpha\right)^2\psi = (k-\alpha)\left(-i\frac{d}{d\phi} - \alpha\right)\psi = (k-\alpha)^2\psi.$

$\Rightarrow \quad H\psi = \frac{\hbar^2}{2I}(k-\alpha)^2\psi.$

$\therefore \quad \boxed{E = \frac{\hbar^2}{2I}(k-\alpha)^2}$

~~Twisted~~ ~~boun~~ B.C.s : $\psi(\phi+2\pi) = \cancel{e^{i2\pi\alpha}}\,\psi(\phi)$

(Since we are including the gauge α in H, the B.Cs are periodic)

$\Rightarrow \quad e^{i(k\cancel{-2\pi\alpha})(\phi+2\pi)} = \cancel{e^{i2\pi\alpha}} \cdot e^{ik\phi}$

$\Rightarrow \quad e^{ik2\pi} = \cancel{e^{i2\pi\alpha}}\ 1 \quad \Rightarrow \quad \boxed{k = m \cancel{+\alpha}} \quad ; \ m \in \mathbb{Z}$

$\therefore \quad \psi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{i(m\cancel{+\alpha})\phi}$

$E_m(\alpha) = \frac{\hbar^2}{2I}(m-\alpha)^2$

For $\phi = \phi_0 \rightarrow \alpha = 1$:

$E_m(1) = \frac{\hbar^2}{2I}(m-1)^2 \quad \cancel{\rightarrow E_0(1) = \frac{\hbar^2}{2I}}$

$\psi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{i(m+1)\phi} \quad \cancel{\rightarrow \psi_0(\phi) = \frac{1}{\sqrt{2\pi}} e}$

$\therefore \quad E_{gs}(1) = 0 \quad [m=1]$

$\psi_{gs}(\phi) = \frac{1}{\sqrt{2\pi}} e^{i2\phi}$ $\quad$ [Ans]

(e) $\phi = \frac{\phi_0}{2} \Rightarrow \alpha = \frac{1}{2}$

$\therefore \quad E_m(\frac{1}{2}) = \frac{\hbar^2}{2I}(m-\frac{1}{2})^2 \quad ; \quad \psi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}$

$\therefore \quad E_{gs}(\frac{1}{2}) = \frac{\hbar^2}{8I} \quad [m = 0, 1]$ $\quad$ [Ans]

$\therefore \quad \psi_{gs}(\phi) = \frac{1}{\sqrt{2\pi}} \quad [m=0]$

$\psi_{gs}(\phi) = \frac{1}{\sqrt{2\pi}} e^{i\phi} \quad [m=1]$

(f)

(8) (i) $N = a^\dagger a$;

$L_z = \hbar(l-N)$; $L_+ = \hbar\sqrt{2l-N}\, a$

$L_- = \hbar\, a^\dagger \sqrt{2l-N}$

$$L_z|n\rangle = \hbar\sqrt{2l-N}\, a|n\rangle$$
$$= \hbar\sqrt{2l-N}\sqrt{n}\,|n-1\rangle$$
$$= \hbar\sqrt{n[2l-(n-1)]}\,|n-1\rangle$$
$$= \hbar\sqrt{n(2l-n+1)}\,|n-1\rangle$$

let $m = l-n \Rightarrow \boxed{n = l-m}$

$$\therefore L_z|n\rangle = \hbar\sqrt{(l-m)(l+m+1)}\,|n-1\rangle \quad \text{— (i)}$$

$$\therefore L_z|n\rangle = \hbar(l-N)|n\rangle = \hbar(l-n)|n\rangle \quad \text{— (ii)}$$

$$L_z|n\rangle = m\hbar|n\rangle \quad \text{— (ii)}$$

$$\therefore L_-|n\rangle = \hbar a^\dagger\sqrt{2l-N}\,|n\rangle$$
$$= \hbar a^\dagger\sqrt{2l-n}\,|n\rangle$$
$$= \hbar\sqrt{2l-n}\sqrt{n+1}\,|n+1\rangle$$
$$= \hbar\sqrt{(2l-n)(n+1)}\,|n+1\rangle$$

$$L_-|n\rangle = \hbar\sqrt{(l-m+1)(l+m)}\,|n+1\rangle \quad \text{— (iii)}$$

$\because$ ~~n ⇒~~ In (i) ; since $n \to n-1$; $m \to m+1$:

$$L_+|l,m\rangle = \hbar\sqrt{(l-m)(l+m+1)}\,|l,m+1\rangle \quad \text{— (iv)}$$
$$L_-|l,m\rangle = \hbar\sqrt{(l+m)(l-m+1)}\,|l,m-1\rangle \quad [n \to m+1 ;\ m \to m-1]$$
$$\text{— (v)}$$

$$\therefore \; L_z(L_\pm |nlm\rangle) = \hbar(m\pm 1)(L_\pm |nlm\rangle) \quad \text{— (vi)}$$

$$\therefore \; L_\pm(L_z |nlm\rangle) = L_\pm(m\hbar |nlm\rangle) = m\hbar(L_\pm |nlm\rangle) \quad \text{— (vii)}$$

From : $[L_z, L_\pm]|nlm\rangle = [\hbar(m\pm 1) - m\hbar] L_\pm |nlm\rangle$

$$= \pm\hbar L_\pm |nlm\rangle$$

$$\Rightarrow \boxed{[L_z, L_\pm] = \pm\hbar L_\pm} \quad \text{— (viii)}$$

$$\therefore \; [L_+, L_-]|nlm\rangle = (L_+L_- - L_-L_+)|nlm\rangle.$$

$$\therefore \; L_+L_-|nlm\rangle = \hbar\sqrt{(l-m+1)(l+m)}\, L_+|m-1\rangle.$$

$$= \hbar^2\sqrt{(l-m+1)^2(l+m)^2}\,|nlm\rangle.$$

$$= (l-m+1)(l+m)\hbar^2|nlm\rangle.$$

$$\therefore \; L_-L_+|nlm\rangle = \hbar\sqrt{(l-m)(l+m+1)}\, L_-|m+1\rangle$$

$$= \hbar^2\sqrt{(l+m+1)^2(l-m)^2}\,|nlm\rangle.$$

$$= \hbar^2(l+m+1)(l-m)|nlm\rangle$$

$$\therefore \; [L_+, L_-]|nlm\rangle = \hbar^2[(l-m+1)(l+m) - (l+m+1)(l-m)]|nlm\rangle$$

$$= \hbar^2[(l-m)(l+m) + (l+m) - (l+m)(l-m) - (l-m)]|nlm\rangle$$

$$= \hbar^2 2m|nlm\rangle$$

$$\therefore \; \boxed{[L_+, L_-] = 2m\hbar^2 = 2L_z\hbar} \quad \text{— (ix)}$$

$$\therefore \; \cancel{L_x = L_+} \quad L_+ = L_x + iL_y$$

$$L_- = L_x - iL_y$$

$$\therefore \; L_x = \frac{L_+ + L_-}{2} \; ; \; L_y = \frac{(L_+ - L_-)}{2i}$$

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$$\therefore [L_z, L_x] = \frac{1}{2}[L_z, L_+ + L_-]$$

$$= \frac{1}{2}([L_z, L_+] + [L_z, L_-])$$

$$= \frac{1}{2}(\hbar L_+ - \hbar L_-) = \frac{\hbar}{2}(L_+ - L_-)$$

$$\boxed{[L_z, L_x] = i\hbar L_y}$$ (Proved).

Similarly $[L_y, L_z] = \frac{1}{2i}[L_+ - L_-, L_z]$

$$= \frac{1}{2i}(-\hbar L_+ - \hbar L_-)$$

$$\boxed{[L_y, L_z] = i\hbar L_x}$$ (Proved)

$$= \frac{-\hbar}{2i} \cdot 2L_x = i\hbar L_x.$$

$$[L_x, L_y] = \left[\frac{L_+ + L_-}{2}, \frac{1}{2i}(L_+ - L_-)\right]$$

$$= \frac{1}{4i}(-[L_+, L_-] + [L_-, L_+])$$

$$= \frac{1}{4i}(-2L_z\hbar - 2L_z\hbar) = -\frac{\hbar L_z}{i} = i\hbar L_z.$$

$$\boxed{[L_x, L_y] = i\hbar L_z}$$ (Proved)

$$L^2 = L_z^2 + L_x^2 + L_y^2$$

$$= L_z^2 + \frac{1}{4}(L_+ + L_-)^2 + -\frac{1}{4}(L_+^2 - L_-^2).$$

$$= L_z^2 + \frac{1}{4}[L_+^2 + L_-^2 + L_+L_- + L_-L_+ - L_+^2 - L_-^2 + L_+L_- + L_-L_+]$$

$$L^2 = L_z^2 + \frac{1}{2}(L_+L_- + L_-L_+)$$

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$$\therefore L^2 |l, m\rangle = \left[m^2\hbar^2 \cancel{|l,m\rangle} + \frac{1}{2}(2m\hbar^2 + 2L_-L_+) \right] |l, m\rangle$$

$$= \left[m^2\hbar^2 + m\hbar^2 + \hbar^2(l+m+1)(l-m) \right] |l, m\rangle$$

$$= \left[\cancel{m^2}\hbar^2 + \cancel{m\hbar^2} + l^2\hbar^2 - \cancel{m^2\hbar^2} + (l-\cancel{m})\hbar^2 \right] |l, m\rangle$$

$$= l(l+1)\hbar^2 |l, m\rangle.$$

$$\Rightarrow \boxed{L^2 = l(l+1)\hbar^2}$$ (Proved)

(b) (ii) In standard angular momentum algebra:

$$L_z |l, m\rangle = m\hbar |l, m\rangle \; ; \quad m = -l \rightarrow +l.$$

In our ~~questi~~ solution:

$$L_z |n\rangle = \hbar(l-n) |n\rangle.$$

$$\therefore m = l - n. \Rightarrow n = l - m$$

$$\therefore m_{max} = l \Rightarrow n_{min} = l - m_{max} = 0$$

$$m_{min} = -l \Rightarrow n_{max} = l - m_{min} = 2l$$

[Ans]